

Where Things Become Imageries Visualization Experience of Chinese Poetic Imageries in Immersive VR

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Figure 1: (a) Conceptual diagram of the project, highlighting the main interaction logic. (b) User position within the VR spatial environment. (c) First-person view of the user.

Abstract

Where Things Become Imageries is a visualization project based on Three Hundred Tang Poems. It constructs a corpus of 4,496 poetic images, from which frequency and semantic richness are extracted to generate parametric “imagery stones.” These are spatially arranged within a semantic field defined by two axes: “subjective/objective” and “abstract/concrete.” Users engage through interactive modes—knocking, bridging, and threading—that activate dynamic particle linkages and reveal corresponding poetic lines. The work creates an immersive poetic network in VR, translating classical poetry from text into spatial and interactive experience.

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1 Introduction

Classical Chinese poetry, especially from the Tang dynasty, is built upon *yixiang* (imagery)—symbolic, layered units that fuse language, perception, and cultural memory. These imageries are nonlinear and ambiguous, resisting conventional computational representation. Most existing visualization approaches focus on rhythm, structure, or surface-level semantics [Cui et al. 2016], with limited capacity to represent metaphorical depth or cultural embeddedness [Drucker 2020]. Immersive systems may offer aesthetic presence, but without semantic grounding, they fail to support interpretive engagement [Khan 2024].

This paper presents a VR-based workflow for visualizing Tang poetic imagery. Based on an existing annotated corpus [Ge 2019] containing 4,496 imagery entries, we propose a new three-category framework—natural, humanistic, and abstract—as the basis for a three-stage

- Parametric modeling of “imagery stones” based on data attributes;
- Semantic spatial layout along abstract/concrete and subjective/objective axes;

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- Audiovisual interaction, using particles to link related imagery and display poetic lines.

By embedding meaning in spatial and embodied forms, the system enables immersive interpretation of classical Chinese poetry.

2 Visualization Design

2.1 Imagery Types and Parameter Mapping

We extract two core metrics—frequency (occurrence rate) and richness (lexical diversity)—to characterize poetic imagery. After normalization, these values drive parametric modeling in Grasshopper, with further refinement and material enhancement in Blender and C4D. Natural imagery takes on soft, irregular forms; humanistic imagery features structured, modular shapes; and abstract imagery appears light and layered, evoking a sense of drift and immateriality.

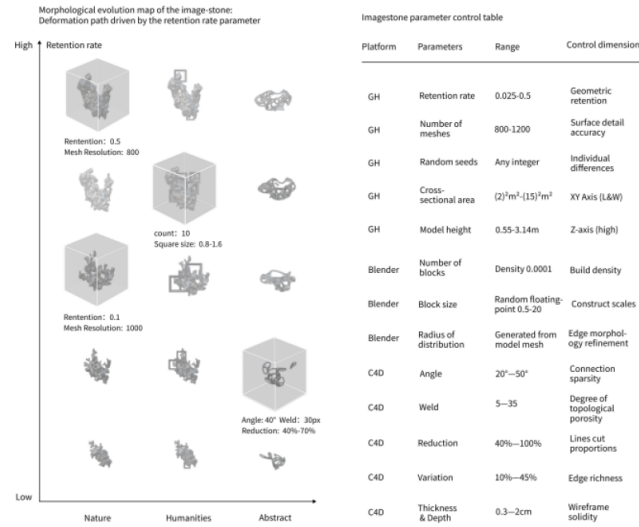


Figure 2: Parametric modeling rules and examples

2.2 Spatial Arrangement and Semantic Axes

We construct a two-dimensional semantic space based on the axes of abstract-concrete and subjective-objective, mapping imagery by cognitive traits. This layout supports spatial organization and reveals deeper poetic relations such as co-occurrence, association, and metaphor.

3 Immersive Interaction System Design

In this system, each “imagery stone” functions as both a semantic modeling unit and an interactive node. Based on the perceptual and cultural attributes of different imagery types, we design three gesture-based interaction modes: tapping for natural imagery, which triggers ambient animations and environmental sounds (e.g., wind, water, falling leaves); lifting and placing for humanistic imagery, activating structural assembly effects with narration and contextual soundscapes; and mid-air pulling for abstract imagery, dissolving the stone into emotional particles and evoking drifting

ambient audio. All interactions also display the imagery’s type, frequency, and related Tang poetry lines in both Chinese and English to support semantic comprehension.

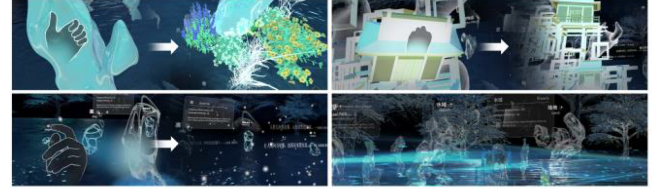


Figure 3: Interactive gestures and corresponding responses

The system also introduces a semantic linkage mechanism. When a stone is activated, dynamic particle lines highlight connections to related imagery and co-occurring Tang poetry lines, forming a cognitive path of evocation, linkage, and resonance. A gesture-based teleport system allows users to shift positions and enter new thematic scenes across semantic islands, creating a spatially guided poetic journey.

4 Conclusion

This paper transforms Tang poetic imagery into a perceptible VR experience through parametric modeling and spatial interaction, enabling semantic exploration and expanding the path of poetic visualization. Future work will incorporate more textual data, AI-generated scenes, and user evaluation to validate its value in cultural education.

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